COMP2850 Software Engineering

Portfolio 1.1

Expectations

Timeliness

You should submit a reasonable attempt at the portfolio task to Gradescope by the deadline provided below. If you are unwell or otherwise unable to complete the work on time, an alternative late submission portal will open after this deadline.

Note that you may only submit **three separate** portfolio tasks late per semester without failing the portfolio.

Once you have submitted a reasonable attempt, you are able to refine and resubmit this work.

Presentation

Follow the instructions provided for the presentation of your work. Follow any instructions provided below regarding file, function or variable names which may be required for the autograder in code tasks.

Integrity

You are responsible for the content of your portfolio, which must consist entirely of your own work. You can find out more about Academic integrity on the Skills@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback

Feedback will be provided through Gradescope within 3 weeks of the submission deadline.

Support opportunities

You can ask for help with your portfolio tasks in any of your timetabled teaching sessions.

Generative AI

In your portfolio you may use generative AI only in a **supportive, tutoring** capacity. You should not be submitting any work which has been produced by AI.

You **CAN** use generative AI to:

- Explain or clarify a topic which you don't fully understand or want extra support with.

You **CANNOT** use generative AI to:

- Produce code, writing or other artefacts which you submit as part of your portfolio work.
- Debug or 'fix' your code for you.

Generative AI may be incorrect - use it at your own risk.

Portfolio 1.1 Deadline **Friday 09/10/2026 1400**

Task – Heron's Formula

Note – you can access these instructions in markdown which shows code more clearly and with syntax highlighting in your Tasks repository.

In src/Main.kt, write a program that uses Heron's formula to compute the area of a triangle, given the lengths of its three sides.

The three side lengths must be provided as command line arguments. You **can** assume that any arguments supplied on the command line are valid floating-point numbers, but you should **not** assume that users always provide exactly three arguments, and must validate this yourself.

If there are fewer than three arguments, your program should use println() to print the *exact* message below:

```
Error: values for a, b, c required on command line
```

After printing this message, your program should terminate immediately, with an exit status code of 1.

If all side lengths have been specified on the command line, your program should compute area and then use println() to print the string "Area = ", followed by the area value. The latter should be formatted to **5 decimal places of accuracy**.

For example, if the program is invoked with

```
./kotlin run 3.0 4.0 5.0
```

then the output should look *exactly* like this:

```
Area = 6.00000
```

Note: do not add anything else to the program or generate any output other than what is specified above! For example, you should not add any code to check whether the side lengths represent a valid triangle.

If you add anything else, or deviate from the instructions above in any way, checking of your solution may fail and you will not be awarded a pass for this portfolio assignment!

Checking Your Solution

On Linux or macOS, you can check whether your solution performs as required by running this command in a terminal window:

```
./kotlin build && python check.py
```

(You may need to use python3 instead of python, depending on where you are doing this and how Python has been installed on that system.)

If you see no failures, your solution is good enough to earn a pass.

Note that this command does not work on Windows systems.

Passing Criteria

In order to pass this portfolio your code must pass **all** automated tests on Gradescope.

Minimum Submission Expectations

A minimum 'reasonable submission' which can be accepted:

- Compiles correctly on Gradescope
- Passes a minimum of 5 automated tests.